

KQL Threat Hunting Lab

Six investigations across authentication, process execution and network activity.

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Azure Data Explorer free cluster | October 2026

367

synthetic events

6

documented hunts

7

dashboard tiles

A match begins an investigation.

The project combines reproducible KQL with contextual triage: matching failed and successful logons, inspecting encoded commands, and linking three network records to a specific process instance.

Fictional, normalized teaching data. Findings demonstrate manual hunting and evidence assessment. Live collection and operational alert deployment are outside this completed scope.

Six hunts, validated outputs

Twelve independently runnable query blocks were validated in ADX on 4-5 October 2026. The table below lists each hunt's headline result.

ID	Hunt	Output	Interpretation
01	Repeated failed logons	2 groups	6 alex failures; 7 labadmin failures
02	Success after failures	1 success	alex succeeds after 6 matching failures
03	Local account creation	2 events	testuser and svc_support
04	Encoded PowerShell	2 events	Both decode to harmless text-output commands
05	Rare process paths	2 pairs	Temp helper and System32 whoami.exe
06	Rare outbound destinations	2 groups	3 helper connections matched to its process start

Dataset baseline

Event family	Meaning	Records
Windows Security 4624	Successful logon	65
Windows Security 4625	Failed logon	37
Windows Security 4720	Account created	3
Sysmon-style 1	Process creation	131
Sysmon-style 3	Network record	131

Snapshot: 24-30 September 2026, UTC. Every record is synthetic. Output units differ across hunts; these totals do not count confirmed attacks.

Following the WIN11-01 sequence

On 30 September 2026, records for alex supply a timeline worth reviewing. The authentication hunt matches computer, target domain, account, source IP and logon type.

Time (UTC)	Recorded observation
09:00:00-09:02:30	Six incorrect-password failures from 10.0.0.50, logon type 10
09:03:00	Successful logon with the same matching fields; 30 seconds after the last failure
09:05:00	alex creates the local svc_support account
09:07:00	PowerShell uses -EncodedCommand; parent image is cmd.exe
09:10:00	Temp-directory sync-helper.exe starts; parent image is PowerShell
09:10:10-09:12:10	Three network records match the helper process identifier

What the sequence establishes

The timestamp order is present in the synthetic records. The helper-to-network link is supported by Computer + ProcessGuid. Missing logon-session identifiers and ParentProcessGuid limit attribution between the earlier events.

Context before escalation

Verify the account owner, source device, expected remote access and account-creation purpose. A separate labadmin group has seven failures without a matching later success. The sam comparison has three failures before success and stays below the five-failure threshold.

ATT&CK; hypotheses: T1110.001 (Password Guessing) and T1136.001 (Local Account). The records do not reveal attempted passwords or establish unauthorized persistence.

Context changes the triage

Encoded PowerShell

Both detected parameters carry Base64/UTF-16LE text. Decoding returns two literal message-output commands, verified in local Windows PowerShell.

Host / user	Parameter	Decoded command
WIN11-02\labadmin	-eNc	Write-Output 'Approved maintenance check'
WIN11-01\alex	-EncodedCommand	Write-Output 'KQL-Lab-Demo'

The maintenance-themed message does not itself verify approval. The alex event follows the earlier logon and account creation, but its decoded contents do not establish malicious execution.

Rare process paths

Finding	Recorded context	Triage
sync-helper.exe	WIN11-01; user Temp path; PowerShell parent; one observation	Higher-priority follow-up because of combined context
whoami.exe /all	WIN11-02; System32 path; cmd.exe parent; one observation	Comparison utility; rarity also captures ordinary tools

Rarity is a lead

A single observation per host/path in a small snapshot is a baseline heuristic. The data contains no helper binary, signature or file hash. Neither an executable name nor a parent image establishes authenticity or intent.

T1059.001 (PowerShell) is relevant to potential misuse being investigated. No ATT&CK; technique is assigned solely from rare execution.

One process, three network records

The focused follow-up joins process creation to outbound network records using **Computer + ProcessGuid**, then checks that each connection is at or after process start.

Matched helper instance

WIN11-01 / WIN11-01\alex
Process start: 30 September 2026, 09:10:00 UTC
Image: C:\Users\alex\AppData\Local\Temp\sync-helper.exe

b8d52208-b0ac-4027-84e5-c8bb576f5035

Connection (UTC)	Destination	Source record
09:10:10	198.51.100.77:4444 / tcp	LAB-000337
09:11:10	198.51.100.77:4444 / tcp	LAB-000338
09:12:10	198.51.100.77:4444 / tcp	LAB-000339

The creation-side parent is PowerShell. The final query names it ProcessParentImage to avoid colliding with the network record's empty ParentImage field.

Observation, then interpretation

The first connection is 10 seconds after process creation. Three records provide only two 60-second intervals. This suggests a question about periodicity; it does not confirm sustained beaconing, an application protocol or command-and-control.

A browser also matches the rarity rule with one TCP connection to 203.0.113.25:443. Both destination addresses are documentation examples under RFC 5737. They are not real malicious indicators.

The join retains matching records only. Missing process starts can exclude relevant traffic, while duplicate starts can multiply matches.

Dashboard and reproducible files

The native ADX dashboard shows the 367-event overview and one table for each hunt. The image is an excerpt of the executed dashboard; complete values remain in the dataset and query files.

Dataset overview (synthetic; timestamps UTC)

TotalEvents	FirstEvent	LastEvent
367	24/09/2026, 07:00:00	30/09/2026, 15:01:00

01 - Repeated failed logons (5+ in 10 min)

Computer	TargetUserName	SourceIp	FailedLogons	WindowStart
WIN-SRV-01	labadmin	10.0.0.60	7	30/09/2026, 11:10:00
WIN11-01	alex	10.0.0.50	6	30/09/2026, 09:00:00

02 - Success after repeated failures

Computer	TargetUserName	SourceIp	LogonType	FailedLogons	SuccessTime
WIN11-01	alex	10.0.0.50	10	6	30/09/2026, 09:00:00

03 - Local account creation

TimeGenerated	Computer	CreatedBy	NewAccount
28/09/2026, 11:30:00	WIN11-02	labadmin	testuser
30/09/2026, 09:05:00	WIN11-01	alex	svc_support

04 - Encoded PowerShell use

TimeGenerated	Computer	User	ParentImage	CommandLine
29/09/2026, 16:10:00	WIN11-02	WIN11-02\labadmin	C:\Windows\explorer.exe	powershell.exe -NoPr
30/09/2026, 09:07:00	WIN11-01	WIN11-01\alex	C:\Windows\System32\cmd.exe	powershell.exe -NoPr

05 - Rare process paths (1 observation)

Computer	Image	Executions	FirstSeen
WIN11-01	C:\Users\alex\AppData\Local\Temp\sync-	1	30/09/2026, 09:10:00
WIN11-02	C:\Windows\System32\whoami.exe	1	30/09/2026, 14:05:00

06 - Rare outbound destinations (3 or fewer)

DestinationIp	DestinationPort	Protocol	Connections	FirstSeen	LastSeen
203.0.113.25	443	tcp	1	30/09/2026, 08:40:00	30/09/2026, 08:40:00
198.51.100.77	4444	tcp	3	30/09/2026, 09:10:10	30/09/2026, 09:10:10

Actual ADX dashboard results, 5 October 2026. Zoom the PDF to inspect the screenshot. The overview has 367 events; hunt tiles 01-06 contain 2, 1, 2, 2, 2 and 2 rows.

Reproduction path

1. Import the CSV once into SecurityLabEvents in an ADX free-cluster database.
2. Verify 367 records, original UTC timestamps, schema and unique source IDs.
3. Run each complete KQL block and compare with the validation matrix.
4. Set your cluster URI in the portable dashboard template, import and save it.

What accompanies this report

The publication pack includes the synthetic CSV, six hunt files, a decoder, six case write-ups, evidence excerpts, the dashboard template, a setup guide and an optional dataset verifier.

Skills and evidence discipline

What this project demonstrates

KQL filtering and aggregation; fixed and rolling time windows; explicit joins; command-line pattern matching; text decoding; identity and process correlation; comparison-case review; dashboard construction; and clear documentation of evidence and limits.

Where the evidence stops

The dataset is fictional and reduced. There is no live endpoint collection, executable sample, DNS or packet evidence, complete parent/session attribution, scheduled Sentinel analytics or operational incident response. The exercise does not estimate production precision, recall or false-positive rates.

Future work

Use richer telemetry to validate exact parent/session links, review file provenance and network contents, test thresholds against a representative labeled baseline, and evaluate operational alert handling. These would be extensions to the completed lab.

Primary sources

Microsoft: Sysmon event and process semantics

Microsoft: PowerShell encoded-command format

Microsoft: KQL inner join

RFC 5737: Documentation IPv4 address blocks

MITRE: T1110.001 - Password Guessing

MITRE: T1136.001 - Local Account

MITRE: T1059.001 - PowerShell

Explore the work

The accompanying project folder contains the complete findings, reproducible queries and source data. The GitHub repository link can be included in the LinkedIn post when the files are published.